# FlyByWire feature user manuals

One manual per `core/*` branch of the fork (36 branches) and one per fork-only `program/*` branch (2): what each feature does, how to use it, how to test it in the sim, what to expect, and its limitations with the reason

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|---|---|
| **Aircraft** | A32NX and A380X |
| **Where to find it** | This folder: each manual as HTML (with its `img/NN/` pictures) and as PDF; `FBW-user-manuals.pdf` holds them all, `FBW-user-manuals.docx` is the editable Word version |
| **Branch** | The `core/*` branches of the fork `TCdr/aircraft`: what to open the upstream pull requests from |
| **Status** | Restructured from the 31 per-PR manuals (their content is merged into the core manuals); each feature carries its own test status. |
| **Pictures** | Instrument pictures come from the local browser harness running the installed instrument bundles; the native WASM displays (weather radar, terrain, VD) are in-sim screenshots; audio features use diagrams. Features added after the per-PR manuals are described in text. |


1. The manuals
2. Program branches (fork-only, not upstream PRs)
3. Stacks: what must be merged first
4. Companion changes in aircraft-large-files
5. How to read them


## 1. The manuals

| # | Manual | Aircraft | Branch | Based on | Features |
|---|---|---|---|---|---|
| 1 | Weather radar and terrain on the ND and VD (PDF) | A32NX and A380X | `core/nd-weather-terrain` | `master` | 8 |
| 2 | Surveillance: SURV controls, ECAM SURV lines and TCAS (PDF) | A380X (SURV panel, VD messages, ECAM lines), A32NX + A380X (TCAS) | `core/surveillance` | `master` | 8 |
| 3 | Independent CPT and F/O MCDUs (PDF) | A32NX | `core/a32nx-mcdu` | `master` | 3 |
| 4 | ECAM procedures, EMER CANC and FIRE TEST (PDF) | A380X (ECAM, FWS) and A32NX + A380X (FIRE TEST option) | `core/ecam-fws` | `master` | 5 |
| 5 | Runway awareness: RAAS and OANS (PDF) | A32NX (RAAS) and A380X (OANS) | `core/oans-btv-raas` | `master` | 5 |
| 6 | A380X standby instruments (ISIS) (PDF) | A380X | `core/isis` | `master` | 4 |
| 7 | Lights: strobe and beacon timing, interior lighting presets (PDF) | A32NX and A380X | `core/lights` | `master` | 3 |

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|---|---|---|---|---|---|
| 8 | Fire warnings and the ENG FIRE pushbutton (PDF) | A32NX (fire warnings, LP fuel valve) and A380X (LP fuel valve) | `core/fire` | `master` | 3 |
| 9 | Electro-hydrostatic actuator accumulator start pressure (PDF) | A32NX and A380X | `core/eha-accumulator-start-pressure` | `master` | 1 |
| 10 | A380X MFD FMS and SURV pages (PDF) | A380X (the shared flight plan code of DIRECT TO, the 50 waypoint limit, the received company winds and the CHECK ALTN WIND check is also in the A32NX FMS; the A32NX keeps its own alternate cruise level and history wind rules; changes the A32NX MCDU too) | `core/a380x-mfd-fms` | `core/surveillance` | 18 |
| 11 | A380X ATC COM and datalink (PDF) | A380X (the ACARS provider connection is shared with the A32NX) | `core/a380x-atccom-datalink` | `core/a380x-mfd-fms` | 7 |
| 12 | A380X fuel jettison and FINAL fuel (PDF) | A380X | `core/fuel` | `core/a380x-mfd-fms` | 7 |
| 13 | GPS receivers, GPS deselection and ECAM STATUS (PDF) | A32NX and A380X (shared systems code); the ECAM STATUS page and STS reminder are A32NX only | `core/gps` | `core/a380x-mfd-fms` | 8 |
| 14 | A380X interactive ND (PDF) | A380X | `core/a380x-interactive-nd` | `core/a380x-mfd-fms` | 2 |
| 15 | flyPad performance calculators (PDF) | A380X and A32NX (one shared flyPad screen per calculator; the company datalink reply time is A380X only) | `core/efb-performance` | `core/a380x-mfd-fms` | 12 |
| 16 | Cockpit printers (PDF) | A380X (printing, buttons) and A32NX (paper door, no torn paper edge) | `core/printer` | `core/a380x-atccom-datalink` | 5 |
| 17 | flyPad ground: taxi route, ATC frequencies and GSX (PDF) | A32NX and A380X (the OANS flags are A380X only) | `core/efb-ground` | `core/efb-performance` | 8 |
| 18 | flyPad Material 3 redesign and independent flyPads (PDF) | A32NX and A380X (the Buffet tab: A32NX only) | `core/efb-redesign` | `core/efb-ground` | 27 |
| 19 | A32NX air conditioning per the FCOM (PDF) | A32NX | `core/a32nx-air-conditioning` | `core/gps` | 4 |

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|---|---|---|---|---|---|
| 20 | A380X cockpit buttons, lights and probes (PDF) | A380X | `core/a380x-cockpit` | `master` | 6 |
| 21 | A32NX exterior: spoilers by panel group and visible RAT (PDF) | A32NX | `core/a32nx-exterior` | `master` | 2 |
| 22 | A380X electrical alerts and AC ESS loads (PDF) | A380X | `core/a380x-elec` | `core/gps` | 6 |
| 23 | A380X flyPad failures (PDF) | A380X | `core/failures-a380x` | `master` + `core/fire` + `core/ecam-fws` | 24 |
| 24 | A32NX flyPad failures (PDF) | A32NX (the fire visual effects, the smoke detection function and the anti-ice systems part also A380X) | `core/failures-a32nx` | `core/gps` + `core/fire` + `core/fcom-audit-fixes` + `core/a32nx-air-conditioning` | 51 |
| 25 | Display unit failures and reconfiguration (PDF) | A32NX and A380X | `core/display-unit-failures` | `core/nd-weather-terrain` | 6 |
| 26 | A380X weather radar GAIN, ELEVN and TILT (PDF) | A380X | `core/a380x-wxr-controls` | `core/nd-weather-terrain` + `core/a380x-mfd-fms` | 5 |
| 27 | A380X VD airports as flat ground (PDF) | A380X (the map view fix: A32NX and A380X) | `core/a380x-vd-runways` | `core/nd-weather-terrain` + `core/oans-btv-raas` | 3 |
| 28 | FCOM audit fixes (PDF) | A32NX and A380X | `core/fcom-audit-fixes` | `master` | 16 |
| 29 | RAT ground stow and A32NX RAT performance (PDF) | A32NX and A380X | `core/rat-ground-stow` | `master` | 2 |
| 30 | A380X cargo doors follow the hydraulics (PDF) | A380X (the full-open fix: A380X and A32NX) | `core/a380x-cargo-doors` | `master` | 2 |
| 31 | A380X ATC mailbox with the SD on an ND display unit (PDF) | A380X | `core/a380x-mailbox-du-reconf` | `core/display-unit-failures` + `core/a380x-atccom-datalink` | 2 |
| 32 | A380X ELEC, fuel and FCOM gap alerts (PDF) | A380X (A32NX: ADIRU power supplies, engine anti-ice valves and anti-ice key events, through shared code) | `core/a380x-elec-fuel-alerts` | `core/a380x-elec` + `core/failures-a380x` + `core/fcom-audit-fixes` | 32 |
| 33 | A380X OIS and ATC datalink system model (PDF) | A380X | `core/a380x-ois-datalink` | `master` + `core/printer` + `core/a380x-mailbox-du-reconf` + `core/a380x-elec` | 8 |

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|---|---|---|---|---|---|
| 34 | A380X OIS: NSS AVNCS and FLT OPS applications (PDF) | A380X | `core/a380x-ois` | `master` + `core/a380x-ois-datalink` + `core/a380x-cockpit` + `core/efb-performance` | 23 |
| 35 | A380X COMPANY MSG memo and voice (PDF) | A380X | `core/a380x-ois-alerts` | `core/a380x-ois` + `core/a380x-elec-fuel-alerts` | 1 |
| 36 | A380X staggered cockpit power-up (PDF) | A380X | `core/a380x-power-up` | `core/a380x-ois` + `core/isis` | 2 |

## 2. Program branches (fork-only, not upstream PRs)

These two branches are cut from `develop` of the fork and are not meant to be opened as upstream pull requests: they hold the long programs of the fork (flyPad failure details, and the FCOM gap fixes with the predictive terrain and obstacle alerting). Their numbers continue the list above.

| # | Manual | Aircraft | Branch | Based on | Features |
|---|---|---|---|---|---|
| 37 | flyPad failure details (PDF) | A32NX and A380X | `program/efb-failure-details` | `develop` `9b2228143` | 6 |
| 38 | FCOM gap fixes (program overview) (PDF) | A32NX and A380X | `program/fcom-gap-fixes` | `develop` `9ad68d612` | 10 |

## 3. Stacks: what must be merged first

Most branches are based directly on `master` (FlyByWire master 66021c244 merged in; `core/a380x-cargo-doors` starts from it) and can be merged in any order. The others are stacked: a branch needs every branch under it merged first.

| Stack | Merge order |
|---|---|
| A380X MFD | `core/surveillance` > `core/a380x-mfd-fms` > then any of `core/a380x-atccom-datalink`, `core/fuel`, `core/gps`, `core/a380x-interactive-nd`, `core/efb-performance` |
| Printer | … `core/a380x-atccom-datalink` > `core/printer` |
| flyPad | … `core/efb-performance` > `core/efb-ground` > `core/efb-redesign` |
| GPS | … `core/gps` > `core/a32nx-air-conditioning`, `core/a380x-elec`, `core/failures-a32nx` (which also merges `core/fire`, `core/fcom-audit-fixes` and `core/a32nx-air-conditioning`: its A32NX FWC alerts build on their code) |
| A380X failures | `core/fire` + `core/ecam-fws` > `core/failures-a380x` (merged in for the engine failures) |
| A380X ELEC, FUEL and FCOM gap alerts | `core/a380x-elec` + `core/failures-a380x` + `core/fcom-audit-fixes` > `core/a380x-elec-fuel-alerts` (the alerts need the ELEC failures of one and the FWS engine and fuel code of the other; its FCOM alert additions also extend the A380X door, A-ICE and NAV alert files of core/fcom-audit-fixes) |
| ATC mailbox | `core/display-unit-failures` + `core/a380x-atccom-datalink` > `core/a380x-mailbox-du-reconf` (the mailbox follows the DU reconfiguration of one and the ATC datalink of the other) |
| ND | `core/nd-weather-terrain` > `core/display-unit-failures`; `core/nd-weather-terrain` + `core/a380x-mfd-fms` > `core/a380x-wxr-controls`; `core/nd-weather-terrain` + `core/oans-btv-raas` > `core/a380x-vd-runways` |

| | |
|---|---|
| Power-up stagger | `core/a380x-ois` + `core/isis` > `core/a380x-power-up` (its OIT and SND changes build on the OIT of one and the SND of the other) |
| Program branches (on develop) | `program/efb-failure-details` (cut from develop `9b2228143`) and `program/fcom-gap-fixes` (cut from develop `9ad68d612`) need `develop`; they are not upstream pull request branches |
| Independent (on master) | `core/a32nx-mcdu`, `core/ecam-fws`, `core/oans-btv-raas`, `core/isis`, `core/lights`, `core/eha-accumulator-start-pressure`, `core/fire`, `core/a380x-cockpit`, `core/a32nx-exterior`, `core/fcom-audit-fixes`, `core/rat-ground-stow`, `core/a380x-cargo-doors` |
| Feature split over two branches | The RAT stow row of `core/efb-redesign` needs `core/rat-ground-stow` (the systems side): merge both for the feature to work; the visible RAT of `core/a32nx-exterior` goes back in only after that ground stow |
| Feature split over two branches (cargo doors) | The cargo door row of `core/efb-redesign` (Ground > Services) reads the inhibition that only `core/a380x-cargo-doors` writes: merge both for the row to show the new state |
| Same panel.cfg section | `core/display-unit-failures` (A32NX `DuMessage` gauge) and `core/efb-redesign` (A32NX F/O flyPad) both add `[VCockpit21]`: whichever is merged second renumbers its section to `[VCockpit22]` |
| Features split over two branches (FCOM alert additions) | A380X smoke detection: the SDF Rust and cargo smoke panel lights are on `core/failures-a32nx`, the SMOKE alerts on `core/a380x-elec-fuel-alerts`. A380X NSS smoke: the NSS zone and the SMOKE light of the NSS MASTER sw on `core/failures-a32nx`, the flyPad failures on `core/a380x-elec-fuel-alerts` (both carry the same failure mapping). Smart FIVE HUNDRED: the flyPad pin option on `core/efb-redesign`, the callout on `core/a380x-elec-fuel-alerts`. A32NX CHECK APPR SELECTION: the MCDU message on `core/gps`, the PFD FMA line on `core/failures-a32nx`. Merge both in each case. When `core/failures-a32nx` and `core/a380x-elec-fuel-alerts` meet: adirs.rs differs only by the IR 10 min alignment (keep it), keep the a320 navigation.rs of core/failures-a32nx; the anti-ice Rust is identical |
| Shared engine failure module | `core/failures-a32nx` and `core/failures-a380x` both carry the same fbw-common engine failure files (engine_failure.rs, ata.ts): whichever is merged second keeps one copy; in `failures/mod.rs` and the A380X `a380_systems_wasm` failure list keep the GPS entries of `core/gps` followed by one ATA 72 block. Both also carry the relight EGT watch of engine_start.rs (the A380X schedule sets 865 °C, the A32NX none) |

## 4. Companion changes in aircraft-large-files

Some features change the 3D cockpit models, which live in the separate FlyByWire repository `aircraft-large-files`. Their companion commits are on the fork `TCdr/aircraft-large-files` and must be merged with the core branch.

| Companion commit | What it adds | Core branch |
|---|---|---|
| `5563e31` | A32NX MCDU screens: both MCDUs on one texture (UV halves) | `core/a32nx-mcdu` |
| `36b4bd9` | A380X pedestal printer paper | `core/printer` |
| `27d6123` | A380X printer door, roller unit and paper slit | `core/printer` |
| `99237fb` | A32NX printer door | `core/printer` |
| `24f7b02` | A380X F/O flyPad screen (independent flyPads) | `core/efb-redesign` |
| `d3060ea` | A32NX F/O flyPad screen (independent flyPads) | `core/efb-redesign` |
| `8c14d21` | A380X printer ON light (green window) | `core/printer` |
| `fa6c5d1` | A380X fuel jettison plume at the outlets (stacked on 8c14d21) | `core/fuel` |

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|---|---|---|
| `b8fb739` | A380X MFP (multi-function probe) vane animations | `core/a380x-cockpit` |
| `b790e8d` | A32NX DU reconfiguration: 17 overlay glasses on the EIS screens, DU_MESSAGE and DU_BLANK materials (stacked on 99237fb and d3060ea) | `core/display-unit-failures` |

## 5. How to read them

- Each manual has one numbered section per **feature** of its branch. The grey line under the feature title gives the aircraft, where to find it and its **status** (sim-tested, or not, with the date).
- **What it is for**: the problem the change solves, in a few sentences.
- **How to use**: numbered steps for normal use.
- **How to test**: a table of steps a tester follows in the sim, each with the expected result. If the sim does something else, it is a bug worth reporting with the branch name.
- **What you should see** tables: the expected behaviour in each situation; **Limitations and why**: what does not work yet, the reason and what to do about it.
- Pictures that do not come from the harness (drawn diagrams, in-sim screenshots taken during development) carry an amber line under their caption: they are not representative of the in-game UI and serve as a reference for the explanation only.